

WILL MCGOWAN

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EDUCATION

University of Strathclyde, Glasgow

Sep 2023 - Jun 2027 (expected)

BA (Hons) Economics

- Average grade to date: 2:1
- **Relevant Modules:** Applied Econometrics; Treasury Management and Derivatives; Advanced Microeconomics

Dissertation | MATLAB

In progress, due 2027

"Does the News Pay? Forecasting European Gas Volatility with MIDAS, Storage and Text Data"

- Testing whether news attention improves forecasts of TTF gas price volatility with mixed-frequency (MIDAS) models built on daily storage, market and news data, including an original Google Trends attention index
- Benchmarking out-of-sample against HAR and AR models with Diebold-Mariano tests before, during and after the crisis, and measuring economic value through the Sharpe ratio of a volatility-targeted TTF futures strategy

Belfast Royal Academy, Belfast

Jun 2023

- A-levels: Mathematics (A), Economics (A), Computer Science (B)

PROJECTS

Wheat Calendar-Spread Strategy | *Agricapital Society*

R

- Built a wheat calendar-spread strategy from futures curve structure and contrarian CFTC managed-money positioning, with every parameter fixed before testing to avoid overfitting
- Weekly backtest (Apr 2025 - Oct 2026): Sharpe 0.76 before costs and 0.28 after, maximum drawdown 3.1%; identified turnover of about 50x notional a year as the main drag on returns

Delta-Hedging Simulator

MATLAB

- Simulated selling a three-month call option and delta-hedging it on 50,000 price paths; hedging four times as often halved the hedging error, in line with theory
- Showed that selling at the wrong volatility earns the Black-Scholes price difference on average, and that with 0.5% trading costs weekly hedging beats daily

Monte Carlo Option Pricing Engine

MATLAB

- Built a vectorised Monte Carlo engine pricing European and Asian options under risk-neutral GBM, benchmarked against Black-Scholes and the closed-form geometric-Asian price
- Cut the standard error of arithmetic-Asian estimates by about 97% at $N = 100,000$ using a geometric-Asian control variate alongside antithetic variates, equivalent to roughly 1,300 times as many plain Monte Carlo paths

Empirical CAPM Test

MATLAB

- Tested CAPM on 704 anomaly portfolios across 56 countries (JKP global factor data), regressing each on its local market excess return
- Rejected zero alpha for 32% of portfolios at the 5% level (32.7% with Newey-West errors), over six times the rate expected by chance, and jointly via GRS in 48 of 52 countries
- Momentum alphas were significant in 37 of 55 countries, size in only 4 of 56

WORK EXPERIENCE

Le Set, Glasgow | *Head Mixologist*

Sep 2026 - Present

- Wrote a new 200-cover venue's 11-drink menu, including 4 original cocktails; trained around 15 bar staff

KONG, Glasgow | *Mixologist and Front of House*

Oct 2024 - Jun 2026

- Analysed sales patterns to refine upselling prompts, raising average customer spend by an estimated 25%; reconciled cash and card takings at the end of each shift

LEADERSHIP AND ACTIVITIES

Strathclyde Agricapital Society | *Co-Founder and Treasurer*

Sep 2025 - Present

- Co-founded Strathclyde's first agri-finance society, grew it to 126 members and published weekly research notes

Strathclyde Poker Society | *Vice-President, then President*

Sep 2025 - Apr 2026

- Ran weekly tournaments for around 60 players on £600 a semester, reconciling every buy-in and prize pool

ADDITIONAL INFORMATION

Technical: R, MATLAB, Python, Excel, Git

Interests: poker, over £40,000 in lifetime cash-game profit, using the Kelly criterion to pick buy-in levels and manage bankroll (minimum 50 buy-ins); chess, 14th at the Irish U19 Chess Championships (2023) and won 21 of 22 games in a simultaneous exhibition against players averaging 1860 FIDE; powerlifting, Northern Ireland U19 champion (2023)